Daniel Ng

516-388-9479 | 71danielng@gmail.com | https://ngdaniel.dev/ | linkedin.com/in/ngdaniel17 | github.com/ng-daniel

Experience

Purdue Envision Center - Rosen Center for Advanced Computing West Lafayette, IN
Software Engineer (Realtime Simulation) Mar 2026 - Present

- Developed 6+ new features for an interactive Unity-based fluid dynamics laboratory exploring high-resolution scientific datasets
- Eliminated 100% of runtime crashes caused by unbounded heap growth by designing a fixed-buffer memory allocation system in C#, resulting in stable memory usage during repeated loading of 100+ MB datasets
- Collaborated with staff engineers to develop CollabXR, an open-source mixed-reality classroom, by improving application stability, developing workflows for creating 3D environments, and planning new features
- Restored reliable asset loading by reverse-engineering legacy asynchronous pipelines and tracing data flow from AWS S3 through Unity runtime systems

Purdue Data Mine Corporate Partners Program (Caterpillar Inc.) West Lafayette, IN
Undergraduate Research Software Engineer Aug 2025 – May 2026

- Led a 4–5 member subteam developing the REST API backend for an AI-powered machine data analysis platform using Python, FastAPI, AWS Bedrock, and Docker
- Architected and implemented an MCP server exposing 5+ custom data analysis and visualization tools, including statistical summaries, histograms, time-series graphs, and geospatial mapping
- Developed an LLM orchestration pipeline to automate multi-step analysis and visualization from natural-language queries

Fallingdani Games New York, NY
Independent Game Developer Jul 2023 – Present

- Developed 11 Unity/C# games both independently and with teams, working across programming, design, art, and audio
- Shipped multiple releases under strict 2-3 day competition deadlines by rapidly prioritizing features and managing scope
- Placed top 3% in enjoyment out of 9.5k entries at GMTK Game Jam 2025 by developing novel mechanics for a twin-stick shooter
- Reached 1,700+ views and 700+ plays across 11 total released projects on itch.io

Projects

Falling Engine - Data-Oriented Game Engine | C++, OpenGL, GLSL, GLFW, CMake 2026

- Built a data-oriented C++ game engine with a sparse-set entity-component system (ECS), asset database, and OpenGL renderer, sustaining 60+ FPS with 36,000+ simulated entities and 32,000+ rendered meshes
- Designed a cache-friendly ECS with paged sparse storage and dense component arrays, enabling constant-time component lookup
- Developed a lazy asset pipeline for dependency-linked glTF models with persistent metadata and stable asset references across imports and reloads
- Decoupled rendering from gameplay systems through a graphics-device abstraction with GPU resource caching for vertex buffers, textures, and shader programs

NYC Service Watch - Data Engineering & Analytics Pipeline | Python, PostgreSQL, dbt, Streamlit, Docker, GitHub Actions 2026

- Built an end-to-end ELT pipeline for NYC service request data with incremental ingestion through Polars, PostgreSQL, and dbt
- Improved warehouse reliability by implementing automated dbt quality tests, layered models, and pre-aggregated analytics marts
- Developed a Streamlit dashboard with sample data and visualizations to validate ingestion and transformed warehouse outputs
- Automated CI validation with GitHub Actions and Docker by provisioning databases, running tests, and verifying dashboard startup

Echoes of Isovios - 2D Roguelike Platformer | Unity Engine, C#, Git 2025

- Developed 85% of the enemy roster, including 17 enemy classes and all 4 boss battles, for a Steam-published game using Unity, C#, finite state machines and animation events
- Refactored enemy architecture to separate shared logic from specialized behaviors, enabling faster prototyping and balancing
- Collaborated with a 20+ member multidisciplinary team using Git/GitHub to design, develop and ship a complete game

Education

Purdue University West Lafayette, IN
Bachelor of Science in Computer Science and Data Science, Graphics Concentration Aug 2024 - May 2028

- Cumulative GPA: 3.83/4.0 (4x Dean's List)
- Relevant Coursework: Operating Systems, Fundamentals of Computer Graphics, Numerical Methods, Systems Programming, Data Structures and Algorithms, Linear Algebra, Multivariable Calculus

Technical Skills

Languages: C++, C, C#, GLSL, Python, SQL, Java, R, HTML/CSS
Frameworks/Libraries: OpenGL, GLM, GLFW, FastAPI, Streamlit, PyTorch, NumPy
Tools/Platforms: Unity, Docker, PostgreSQL, Git/GitHub, GitHub Actions, CMake, AWS, Ollama